Review of thermodynamics

Exercise problems

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Problem 1

Problem 1: Consider the flow through a rocket engine nozzle. Assume that the gas flow through the nozzle is an isentropic expansion of a calorically perfect gas. In the combustion chamber, the gas which results from the combustion of the rocket fuel and oxidizer is at a pressure and temperature of 15 atm and 2500 K, respectively; the molecular weight and specific heat at constant pressure of the combustion gas are 12 and 4157 J/kg-K, respectively. The gas expands to supersonic speed through the nozzle, with a temperature of 1350 K at the nozzle exit. Calculate the pressure at the exit.

Given: isentropic flow through the nozzle.

$p_1 = 15\,\text{atm}$, $T_1 = 2500\,\text{K}$

$\hat{m} = 12\,\text{g/mol}$, $c_p = 4157\,\text{J/kg-K}$

$T_2 = 1350\,\text{K}$

$p_2 = ?$

Problem 1 - solution

Given: isentropic flow through the nozzle.
$p_1 = 15\,\text{atm},\ T_1 = 2500\,\text{K}$
$\hat{m} = 12\,\text{g/mol},\ c_p = 4157\,\text{J/kg-K}$
$T_2 = 1350\,\text{K}$
$p_2 = ?$
Solution:
The gas constant can be computed as,

$$R = \frac{R_u}{\hat{m}} = \frac{8314}{12} = 692.833\,\text{J/kg-K}\,.$$

Using the equation for c_p, we can compute the value of γ,

$$c_p = \frac{\gamma R}{\gamma - 1} \quad \Longrightarrow \quad \gamma = \frac{c_p}{c_p - R}$$

$$\gamma = \frac{4157}{4157 - 692.833} = 1.2$$

From isentropic relations,

$$\frac{p_2}{p_1} = \left(\frac{T_2}{T_1}\right)^{\gamma/(\gamma-1)}$$

$$p_2 = p_1 \left(\frac{T_2}{T_1}\right)^{\gamma/(\gamma-1)}$$

$$p_2 = 15 \times \left(\frac{1350}{2500}\right)^{1.2/(1.2-1)}$$

$$\boxed{p_2 = 0.371924\,\text{atm}}\,.$$

Problem 2

Problem 2: Calculate the isentropic compressibility for air at a pressure of 0.5 atm. Compare the result with that for the isothermal compressibility.

Given:

$p = 0.5\,\text{atm}$

$\tau_s =?$

$\tau_T =?$

Problem 2 - solution

Solution:
For an isentropic process,

$$\frac{p_2}{p_1} = \left(\frac{\rho_2}{\rho_1}\right)^{\gamma}$$

which can also be written as,

$$p = c\,\rho^{\gamma} \quad \Longrightarrow \quad \rho = \left(\frac{p}{c}\right)^{1/\gamma}$$

$$\left(\frac{1}{c}\right)^{1/\gamma} = \frac{\rho}{p^{1/\gamma}}$$

with a constant c. Taking a partial derivative of ρ w.r.t. p,

$$\left(\frac{\partial\rho}{\partial p}\right)_s = \left(\frac{1}{c}\right)^{1/\gamma} \frac{1}{\gamma}\, p^{(1/\gamma)-1} = \frac{\rho}{\gamma\, p}$$

The isentropic compressibility can be calculated as,

$$\tau_s = \frac{1}{\rho}\left(\frac{\partial\rho}{\partial p}\right)_s = \frac{1}{\rho}\,\frac{\rho}{\gamma\, p}$$

$$\boxed{\tau_s = \frac{1}{\gamma\, p}}\,.$$

At a pressure of $0.5\,\text{atm} = 0.5 \times 101325 = 50662.5\,\text{Pa}$, the isentropic compressibility of air is,

$$\tau_s = \frac{1}{1.4 \times 50662.5}$$

$$\boxed{\tau_s = 1.40989 \times 10^{-5}\,\text{m}^2/\text{N}}\,.$$

Problem 2 - solution

The isothermal compressibility can be calculated as,

$$\tau_T = \frac{1}{\rho}\left(\frac{\partial \rho}{\partial p}\right)_T$$

The density can be written using the ideal gas equation of state,

$$\rho = \frac{p}{R\,T}$$

and the partial derivative w.r.t. p at constant temperature is,

$$\left(\frac{\partial \rho}{\partial p}\right)_T = \frac{1}{R\,T} = \frac{\rho}{p}$$

The isothermal compressibility is therefore,

$$\tau_T = \frac{1}{\rho}\left(\frac{\partial \rho}{\partial p}\right)_T = \frac{1}{\rho}\,\frac{\rho}{p}$$

$$\boxed{\tau_T = \frac{1}{p}}\,.$$

At a pressure of 50662.5 Pa, the isentropic compressibility of air is,

$$\tau_T = \frac{1}{p} = \frac{1}{50662.5}$$

$$\boxed{\tau_T = 1.97385 \times 10^{-5}\,\mathrm{m}^2/\mathrm{N}}\,.$$